No Answer Needed: Predicting LLM Answer Accuracy from Question-Only Linear Probes

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Introduction

- Do large language models internally anticipate whether they will answer a factual question correctly *before* they generate any answer tokens?
- We test the **Linear Representation Hypothesis** for factual correctness: after a question is read, is there a simple direction in activation space that separates future right from wrong answers?
- Across six open models from 7B to 70B parameters, we find strong evidence that such a question-only direction exists.

Materials and methods

1. Read the prompt, stop at the final question token, and extract the residual stream before generation.
2. Train a one-dimensional difference-of-means direction on TriviaQA. This deliberate simplicity tests linear separability rather than maximizing predictive accuracy with a richer classifier.
3. Compare the probe against black-box assessors and verbalized confidence in distribution, across factual OOD datasets, and on arithmetic GSM8K.

$$s(x) = \langle h(x), v \rangle$$

The score is computed from question-only activations: a positive projection means the internal state already looks more like questions the model will answer correctly.

Results

One simple question-only direction generalizes across factual datasets, beats black-box assessors under domain shift, and supports the Linear Representation Hypothesis for factual correctness.

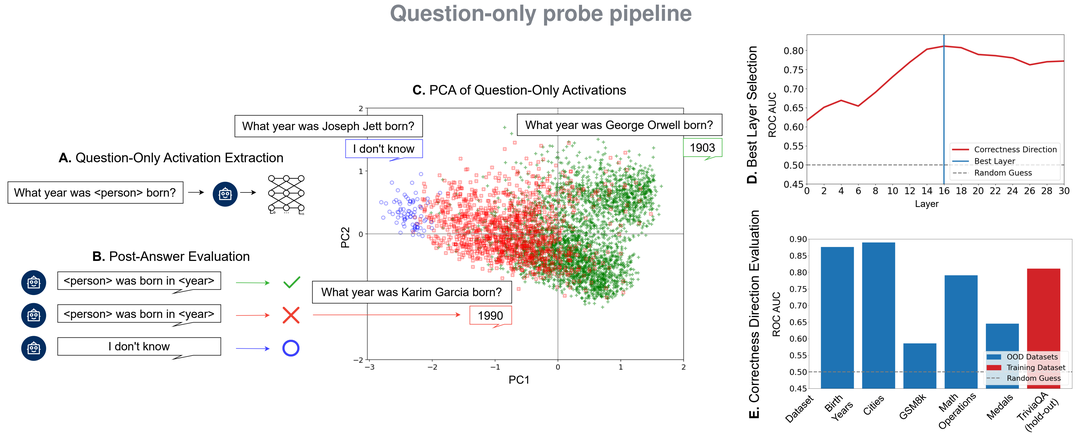
24/30
OOD wins vs best black-box baseline

17/18
factual OOD wins

0.729
mean OOD AUROC

0.752
mean factual OOD AUROC

Question-only probe pipeline



The score is defined from question-only activations before answer generation, then reused across held-out questions and datasets. Assessor averages are lower at 0.594 overall OOD and 0.651 factual OOD.

Sample efficiency is unusually high: about 160 questions already help, and roughly 2,560 recover the full 48,540-question training-set performance.

Conclusions

- These results validate the **Linear Representation Hypothesis** for factual correctness: a one-dimensional pre-answer direction already separates future right from wrong answers.
- The linear probe outperforms black-box assessors under factual domain shift, even though the best assessor remains slightly stronger in distribution.
- The signal emerges only after mid-layer processing, saturates later in the stack, and becomes cleaner in larger models.
- Unprompted "I don't know" responses occupy the negative extreme of the same direction, indicating a latent confidence or knowledge-awareness axis.
- Manual inspection suggests the scores are meaningfully calibrated: high-score mistakes are often near misses rather than arbitrary failures.

For factual QA, the direction behaves like both a self-correctness signal and a self-awareness signal.

Models and data

Six open models from 7B to 70B. Train on TriviaQA 60K and test on Cities, Notable People, Medals, Math Operations, and GSM8K.

Main takeaway

A simple question-only linear probe exposes a factual self-correctness signal before generation, but arithmetic reasoning appears to rely on a different internal mechanism.

Further information

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